

Adversarial Robustness Evaluation and Hardening of an AI-Based Intrusion Detection System for IoT Networks

Project Proposal

Supervisor

Anam Zahid

Submitted by

Zainab Salman [BSAI23024]
Muhammad Luqman [BSAI23054]
Haseeb Farooq [BSAI23003]

Department of ARTIFICIAL INTELLIGENCE
Information Technology University

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1. Introduction and Background

Machine learning has become the default approach for modern Network Intrusion Detection Systems (NIDS), replacing brittle signature-based rules with classifiers that learn attack patterns directly from network traffic. Reported detection accuracies in academic literature frequently exceed 95–99% on standard benchmark datasets. However, these figures are almost always measured under benign, non-adversarial test conditions, where the test data is drawn from the same distribution as the training data. In a real deployment, a motivated attacker does not submit traffic that matches this distribution; they actively probe and adjust their traffic to evade detection while keeping the underlying attack fully functional.

This project studies that gap directly, using Internet of Things (IoT) network security as its setting. IoT devices are increasingly deployed in homes, industrial systems, and critical infrastructure, and organisations are adopting AI-based intrusion detection for these networks without first testing whether that detection holds up against a deliberately evasive attacker. Rather than building another classifier and reporting a single benchmark accuracy figure, this project builds a deep learning-based intrusion detection system for IoT network traffic, evaluates how it behaves when an attacker deliberately disguises attack traffic to evade detection, and then hardens the system against that specific failure mode through adversarial training.

2. Problem Description and Project Objective

The overwhelming majority of AI/ML-based intrusion detection research, and the majority of final-year projects in this area, evaluate models purely on a held-out test split drawn from the same distribution as the training data. This produces an optimistic and somewhat misleading picture of real-world reliability. Adversarial machine learning research has repeatedly shown that small, carefully crafted perturbations to input features can cause a high-accuracy classifier to misclassify malicious traffic as benign, without changing the underlying attack's effectiveness. This has been extensively studied for unstructured data such as images, but its equivalent for structured, tabular network-traffic data is comparatively less mature, and very few studies examine intrusion detection specifically while also proposing and validating a working defense [1], [2]. A further complication is that many published adversarial attacks against intrusion detection systems have not been checked for real-world validity: a recent review found that up to 80.3% of published adversarial examples become invalid once real-world protocol compliance and semantic-preservation constraints are applied, meaning the perturbed traffic could not actually be sent over a real network while still functioning as the intended attack [3]. There is a clear need for a project that (a) generates disguised attacks that respect these real-world constraints, (b) quantifies how much this genuinely degrades a working IDS, and (c) proposes and validates a defense against it.

The project objective, stated concisely, is: to design and train a deep learning-based intrusion detection system for IoT network traffic, evaluate its robustness against protocol-realistic adversarial evasion attacks, and validate an adversarial-

training-based defense that measurably recovers detection performance under these constrained, realistic conditions.

3. Application Areas and Related Work

IoT and IIoT devices are increasingly embedded in homes, industrial control systems, telecommunications infrastructure, and smart-city deployments, and organisations in these sectors are adopting AI-based security tooling without first testing whether it is robust to a motivated, adaptive attacker. A project that quantifies this vulnerability and validates a working defense is directly useful to any organisation deploying AI-based network defense, since it produces a concrete, repeatable methodology for auditing whether such a system will hold up under realistic evasion attempts before it is relied upon in production, rather than a single benchmark accuracy figure.

A 2024 survey on adversarial challenges in network intrusion detection notes that although adversarial threats have been extensively studied for unstructured data such as images and text, their impact on structured data such as network traffic remains comparatively unexplored, and that very few studies have investigated intrusion detection specifically while proposing corresponding defenses [1]. A 2023 survey covering adversarial attack and defense strategies for intrusion detection systems more broadly reaches a similar conclusion, explicitly identifying open gaps and calling for further research into defense and retraining approaches [2]. A more recent 2026 review goes further, applying a constraint-aware framework to a large body of published attacks and finding that protocol compliance and semantic-preservation requirements render up to 80.3% of them invalid in practice; the same review reports that existing defenses, including adversarial training, ensemble methods, and moving target defense, face scalability barriers that limit real production deployment [3]. Taken together, this literature indicates that adversarial robustness testing for intrusion detection is markedly less mature than the equivalent work on images, and that realistic, constraint-aware evaluation of defenses in particular remains limited.

Three papers are especially relevant to this project's specific approach. Kumar et al. (2025) propose a deep autoencoder that detects functionally-preserved adversarial examples by learning the normal statistical pattern of a traffic sample's non-functional attributes, that is, the surface-level features an attacker can alter without breaking the attack's underlying functionality; traffic whose non-functional attributes deviate from this learned pattern is flagged as disguised [4]. Their defense is detection-based: suspicious traffic is flagged and rejected, rather than the classifier itself being retrained to recognise it correctly. Mohammadian et al. (2023) propose a gradient-based attack that uses a Jacobian Saliency Map to select the smallest, most effective subset of features to perturb in order to evade a deep learning-based NIDS, and demonstrate this across the CIC-IDS2017, CIC-IDS2018, and CIC-DDoS2019 datasets, achieving high evasion success while

altering fewer features than competing methods [5]. Neither of these two papers, individually, retrains a classifier and measures its recovery in detection accuracy.

The closest existing work to this project is Vitorino, Praça, and Maia (2023), who generate realistic, constraint-respecting adversarial examples using their Adaptive Perturbation Pattern Method (A2PM) and evaluate four tree-based models, Random Forest, XGBoost, LightGBM, and Isolation Forest, in both a normally-trained and an adversarially-trained configuration, across the IoT-23 and Bot-IoT datasets [6]. Their results show detection accuracy dropping as low as 26–34% under attack in some configurations, and recovering to between roughly 97% and 99.98% after adversarial retraining. This confirms that the overall build-attack-harden pattern this project follows is an established, validated methodology, not a novel invention, and it would be inaccurate to claim otherwise. What this specific paper does not cover is deep learning-based classifiers, since its evaluation is limited to tree-based ensembles, and it does not use the ToN_IoT dataset. This project's contribution is therefore not the invention of a new defense concept, but a focused, honestly-scoped extension: testing whether this same validated pattern, realistic constrained attack generation followed by adversarial retraining, holds up on a deep learning-based classifier and on the ToN_IoT dataset specifically, using an attack-generation method built by combining the non-functional feature framing of Kumar et al. [4] with the efficient gradient-based feature selection of Mohammadian et al. [5].

4. Methodology and Scope

This project uses the ToN_IoT dataset, a labelled collection of IoT and IIoT network traffic developed by Moustafa et al. at UNSW Canberra, which includes raw packet captures (pcaps) alongside telemetry and operating-system-level data spanning both benign operation and a range of attack scenarios [7], [8]. Flow-level features will be extracted from the raw packet captures for use in classifier training, consistent with how the dataset has been processed in prior published work.

A deep learning classifier (a convolutional or recurrent architecture, to be finalised after initial experimentation) will be trained on the preprocessed and normalised traffic features, with standard evaluation metrics, accuracy, precision, recall, F1-score, and ROC-AUC, reported to establish a credible clean baseline before any adversarial testing takes place.

Disguised, constrained adversarial examples will then be generated in two stages: first, traffic features will be separated into functional attributes, which cannot be altered without breaking the attack's underlying effectiveness, and non-functional attributes, which can be altered while preserving that effectiveness, following the framing introduced by Kumar et al. [4]; second, a gradient-based, Jacobian-Saliency-Map-style method, adapted from Mohammadian et al. [5], will be used to select the smallest and most effective subset of those non-functional attributes to perturb. Every generated example will additionally be checked against the feature-dependency relationships present in real network traffic, so that perturbations

remain internally consistent rather than describing traffic that could not actually occur on a real network.

The baseline classifier will then be retrained on a mixture of clean and adversarially-generated examples (adversarial training), and re-evaluated on the same disguised traffic used to degrade the original model. The project's central result is a three-condition comparison: (1) the baseline model on clean, held-out test data, (2) the same baseline model under adversarial attack, and (3) the hardened, retrained model under the same attack, reported using the same accuracy, precision, recall, and F1-score metrics throughout, together with the false-negative rate, since a missed attack is the more costly error type for an intrusion detection system.

The project's scope is deliberately bounded as follows. All experiments are conducted on the publicly available ToN_IoT captures and on traffic replayed from them; the project does not attempt to intercept, defend, or test against any live production network. The attack-generation method is limited to the gradient-based, feature-selection approach described above, combined with functional/non-functional feature filtering; a full comparative benchmark against every published attack family, including GAN-based or full constrained-optimisation-based methods, is outside the project's scope, though these are discussed as possible future extensions. The live dashboard built for the final demonstration is a visualisation and demonstration tool, not a production-grade, real-time packet-per-second network sensor. Finally, the project's findings are scoped to the device types and attack categories represented in the ToN_IoT dataset, and are not claimed to generalise automatically to network types or protocols outside that dataset.

5. Tools and Technology

The project will be implemented primarily in Python, using PyTorch or TensorFlow for deep learning model development, and scikit-learn for baseline comparisons and standard evaluation metrics. Pandas and NumPy will handle data preprocessing. A flow-based feature extraction tool (such as a Zeek- or CICFlowMeter-equivalent flow exporter) will be used to convert the raw ToN_IoT packet captures into flow-level features, and Scapy will be used where packet-level inspection or manipulation is required to validate the realism of generated adversarial traffic. Streamlit will be used to build the live demonstration dashboard. During the literature review process, ResearchRabbit was used to trace related and citing work around the project's core papers, and the CORE Conference and Journal Ranking Portal was used to sense-check the standing of the venues being cited.

All three team members are enrolled in the BS Artificial Intelligence programme and have completed the relevant coursework in machine learning and deep learning ahead of this project; the security-specific components, adversarial machine learning and network traffic analysis, represent a natural extension of this background rather than an entirely new domain, and the team is prepared to build the additional networking and security-specific knowledge the project requires as it

progresses. The project is of shared interest to all three members, who collectively proposed and refined its scope.

6. Milestones

The table below outlines the major milestones for this project across a 24-week timeline.

Activity	Tentative Date
Literature review, dataset and gap finalisation	Weeks 1–3
Dataset acquisition and preprocessing (ToN IoT flow extraction)	Weeks 4–5
Baseline deep learning IDS development and evaluation	Weeks 6–9
Constrained adversarial attack module development	Weeks 10–13
Adversarial impact evaluation (accuracy degradation)	Weeks 14–15
Adversarial training / hardening implementation	Weeks 16–19
Post-defense evaluation and comparative analysis	Weeks 20–21
Live dashboard development and integration	Weeks 22–23
Final report, poster, and defense preparation	Week 24

References

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